# Shuchang Xu

Hong Kong, China  sxuby@connect.ust.hk  shuchangxu.github.io

## Education

**Hong Kong University of Science and Technology** — Hong Kong, China
Ph.D. Candidate in Computer Science, GPA: 4.2/4.3, Advisor: Prof. Huamin Qu — 08/2024 - Present

**Massachusetts Institute of Technology** — Cambridge, USA
Visiting Researcher at MIT Media Lab, Advisor: Prof. Pattie Maes — 04/2026 - 09/2026

**Tsinghua University** — Beijing, China
M.S. in Computer Science, GPA: 3.7/4.0, Advisor: Prof. Yuanchun Shi — 08/2018 - 06/2021

**Tsinghua University** — Beijing, China
B.S. in Electrical Engineering, GPA: 91.2/100 (Top 5%) — 08/2014 - 06/2018

## Best Paper Awards

**UIST 2026 Best Paper Award** — 11/2026
Among the top 1% out of 1,259 submissions

**UIST 2025 Best Paper Honorable Mention Award** — 09/2025
Among the top 3% out of 952 submissions

**CHI 2025 Best Paper Honorable Mention Award** — 04/2025
Among the top 5% out of 5,020 submissions

## Fellowships & Honors

**Hong Kong Ph.D. Fellowship** — 08/2024
Full stipend, approximately USD 150,000 in total

**HKUST RedBird Academic Excellence Award** — 05/2026
HKD 60,000 for academic and research excellence

**Hong Kong JEBN Scholarship for Talents** — 05/2026
HKD 48,000 for academic research support

**HKUST Global Research Award** — 04/2026
HKD 60,000 for global research programs

**HKUST Research Travel Grants** — 01/2026
HKD 90,000 in total for conference travels

**Outstanding Graduate at Tsinghua University** — 06/2021

**China National Scholarship for Postgraduates** (Top 1%) — 11/2020

**China National Scholarship for Postgraduates** (Top 1%) — 11/2019

**China National Scholarship for Undergraduates** (Top 1%) — 11/2017

## Community Services

**CHI 2027 Associate Chair** — 09/2026

**UIST 2026 Workshop Organizer** — 11/2026

**CHI 2024 Late-Breaking Work Associate Chair** — 01/2024

**Special Recognition for Outstanding Reviews**
CHI 2026, UIST 2026, CHI 2025 (×3), IMWUT 2025, ISMAR 2025, and DIS 2026

## Publications

[1] **Sonic Stage: Automatically Generating Interactive Spatial Soundscapes to Facilitate Dialogue Video Comprehension for Blind Viewers.**
**Shuchang Xu**, Xiaofu Jin, Gaurav Jain, Wenshuo Zhang, Huamin Qu, Brian A. Smith, Yukang Yan.
*UIST 2026 - ACM Symposium on User Interface Software and Technology (**Best Paper Award** 🏆).*

[2] **RemiAssist: A Therapist-Supporting System for Photo-Based Reminiscence Therapy in Dementia Care.**
**Shuchang Xu**, Minglong Tang, Junyan Mao, Xiaofu Jin, Wazeer Zulfikar, Yasith Samaradivakara, Jiayi Zhou, Huamin Qu, Yuling Sun, Pattie Maes.
*UIST 2026 - ACM Symposium on User Interface Software and Technology.*

[3] **Branch Explorer: Leveraging Branching Narratives to Support Interactive 360° Video Viewing for Blind and Low Vision Users.**
**Shuchang Xu**, Xiaofu Jin, Wenshuo Zhang, Huamin Qu, Yukang Yan.
*UIST 2025 - ACM Symposium on User Interface Software and Technology.*

[4] **DanmuA11y: Making Time-Synced On-Screen Video Comments (Danmu) Accessible to Blind and Low Vision Users via Multi-Viewer Audio Discussions.**
**Shuchang Xu**, Xiaofu Jin, Huamin Qu, Yukang Yan.
*CHI 2025 - ACM Conference on Human Factors in Computing Systems (**Honorable Mention** 🏅).*

[5] **Memory Reviver: Supporting Photo-Collection Reminiscence for People with Visual Impairment via a Proactive Chatbot.**
**Shuchang Xu**, Chang Chen, Zichen Liu, Xiaofu Jin, Linping Yuan, Yukang Yan, Huamin Qu.
*UIST 2024 - ACM Symposium on User Interface Software and Technology.*

[6] **RegulAR: Graph-Grounded Error Recognition and Assistance for Procedural Tasks in Augmented Reality.**
Yilin Ye, Jindu Wang, Hiu Tung Wong, **Shuchang Xu**, Huamin Qu, Wong Kam-Kwai.
*UIST 2026 - ACM Symposium on User Interface Software and Technology.*

[7] **NeuroSync: Intent-Aware Code-Based Problem Solving via Direct LLM Understanding Modification.**
Wenshuo Zhang, Leixian Shen, **Shuchang Xu**, Jindu Wang, Jian Zhao, Huamin Qu, Linping Yuan.
*UIST 2025 - ACM Symposium on User Interface Software and Technology (**Honorable Mention** 🏅).*

[8] **RhythmTA: a Visual-Aided Interactive System for ESL Rhythm Training via Dubbing Practice.**
Chang Chen, Sicheng Song, **Shuchang Xu**, Zhicheng Li, Huamin Qu, Yanna Lin.
*UIST 2025 - ACM Symposium on User Interface Software and Technology.*

[9] **Wearable AR for Restorative Breaks: How Interactive Narrative Experiences Support Relaxation for Young Adults**
Jindu Wang, Runze Cai, **Shuchang Xu**, Tianrui Hu, Huamin Qu, Shengdong Zhao, Linping Yuan.
*CHI 2026 - ACM Conference on Human Factors in Computing Systems.*

[10] **Tactile Compass: Enabling Visually Impaired People to Follow a Path with Continuous Directional Feedback.**
Guanhong Liu, Tianyu Yu, Chun Yu, Haiqing Xu, **Shuchang Xu**, Ciyuan Yang, Feng Wang, Haipeng Mi, Yuanchun Shi.
*CHI 2021 - ACM Conference on Human Factors in Computing Systems.*

[11] **LightGuide: Directing Visually Impaired People along a Path Using Light Cues.**
Ciyuan Yang, **Shuchang Xu (co-first)**, Tianyu Yu, Guanhong Liu, Chun Yu, Yuanchun Shi.
*IMWUT 2021 - Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.*

[12] **Virtual Paving: Rendering a Smooth Path for People with Visual Impairment through Vibrotactile and Audio Feedback.**
**Shuchang Xu**, Ciyuan Yang, Wenhao Ge, Chun Yu, Yuanchun Shi.
*IMWUT 2020 - Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.*

[13] **Accurate and Low-Latency Sensing of Touch Contact on Any Surface with Finger-Worn IMU Sensor.**
Yizheng Gu, Chun Yu, Zhipeng Li, Weiqi Li, **Shuchang Xu**, Xiaoying Wei, Yuanchun Shi.
*UIST 2019 - ACM Symposium on User Interface Software and Technology.*

[14] **A Machine Learning Framework to Identify Routing Short Violations from a Placed Netlist.**
Aysa Tabrizi, Nima Darav, **Shuchang Xu**, Logan Rakai, Ismail Bustany, Andrew Kennings, Laleh Behjat.
*DAC 2018 - ACM/IEEE Design Automation Conference.*

## Workshop & Poster Publications

[a] **Cyber-Physical Systems for Accessibility and Ability Augmentation: Bridging Diverse Communities.**
**Shuchang Xu**, Riku Arakawa, Mina Huh, Nandi Zhang, Tianyu Zhang, Wazeer Zulfikar, Ruei-Che Chang, Yotam Sechayk, Huamin Qu, Amy Pavel, Franklin Mingzhe Li, Yukang Yan, Brian A. Smith, Pattie Maes
*UIST 2026 Workshop - ACM Symposium on User Interface Software and Technology.*

[b] **From Audio Description to Movement: Challenges and Design Principles in Video-based Body Exercise for Blind and Low Vision Users.**
Ziyi Jiang, Mengjie Jian, Jiajia Hu, Hongze Zhao, Huamin Qu, **Shuchang Xu**, Guanhong Liu.
*CSCW 2025 Companion - ACM Conference on Computer-Supported Cooperative Work and Social Computing.*

[c] **MemeSpeak: Making Image Memes Blind-Accessible through Emotional Speech Augmentation.**
Gengyu Chen, Chiqian Xu, Junru Jiang, Yuting Lu, Xiao Wang, Huamin Qu, **Shuchang Xu**, Guanhong Liu.
*CSCW 2025 Companion - ACM Conference on Computer-Supported Cooperative Work and Social Computing.*

## Academic Talks

**University of Macau**, Invited by Prof. Ziwei Wu 09/2026
AI-Powered Assistive Systems for Enhancing Human Cognition

**University of California, Berkeley, EECS**, Invited by Prof. Amy Pavel 09/2026
AI-Powered Assistive Systems for Enhancing Human Cognition

**Columbia University, CEAL Lab**, Invited by Prof. Brian A. Smith 05/2026
AI-Powered Assistive Systems for Enhancing Human Cognition

**Northeastern University, Ghost Lab**, Invited by Prof. Casper Harteveld 05/2026
AI-Powered Assistive Systems for Enhancing Human Cognition

**KAIST, NC Lab**, Invited by Prof. Junehwa Song 10/2025
AI-Powered Interactive Systems for Improving Visual Media Accessibility

**UIUC, SALT Lab**, Invited by Prof. Yun Huang 06/2025
AI-Powered Interactive Systems for Improving Visual Media Accessibility

## Teaching Experiences

| | |
|---|---|
| **Teaching Assistant**, Java Programming Language, HKUST | 12/2025 - 09/2025 |
| **Teaching Assistant**, C Programming Language, HKUST | 05/2025 - 02/2025 |
| **Guest Lecturer**, User Experience Design, Tongji University | 05/2025 - 02/2025 |
| **Teaching Assistant**, User Experience Design, Tsinghua University | 06/2020 - 02/2020 |
| **Teaching Assistant**, Big Data and Machine Intelligence, Tsinghua University | 01/2020 - 09/2019 |

## Professional Experiences

| | |
|---|---|
| **Hong Kong University of Science and Technology** – **Research Assistant** | 08/2023 - 07/2024 |
| **Xiaomi Corporation** – **Software Product Manager** | 08/2021 - 06/2023 |
| **Bytedance Corporation** – **TikTok Technical Artist** (Intern) | 06/2020 - 08/2020 |
| **Tencent Corporation** – **Game Level Designer** (Intern) | 06/2019 - 08/2019 |
| **NetEase Corporation** – **Game Interaction Designer** (Intern) | 06/2018 - 08/2018 |

## Patents

**Education Technologies**: IP.KH.12803

**Vision Accessibility Techniques**: CN111494175

**Mobile Game Interaction Techniques**: CN109316745, CN109460179, CN109550242

**Smartphone Interaction Techniques**: CN115857375, CN115755635